

AXERAI

— *Products receive digital life*

Full Technical *Reference*

Implementation bible — all modules, flows, env vars, APIs, ledger rules, and edge cases. Companion to the Overview PDF.

22 Sections

Complete File Map

Env + API Spec

Verify + Ledger Deep Dive

Example Code AXE-327T5

DOCUMENT TYPE

Technical FULL

VOICE (DOC)

Inworld API

VERSION

July 2026 Pilot

PAIR WITH

AXERAI-MVP-Document.pdf

IMPLEMENTATION MAP

Intro + Preload

MindAR + Layer A

Verify B + C

Gemini + Inworld

Ledger Memory

Admin Dashboard

App.jsx orchestrates · 25+ src modules

§

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Note: This document uses generic MVP naming (D3 character, code `AXE-327T5`, Inworld voice). Codebase module names (myra*, elevenLabsTts.js) are cited as implementation paths only.

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What This MVP Is

Browser-based, app-less WebAR platform turning physical packaging into a conversational digital touchpoint with persistent per-code memory.

- User opens URL → intro preload → camera grant → MindAR tracks image target on box.
- **Layer A (fast):** product visible → 3D persona appears immediately (Stage 1).
- **Layers B + C (background):** Gemini vision verifies brand/code on packaging + liveness anti-spoof → true/false stored.
- Live mic STT → Gemini reply → Inworld TTS → D3 persona talks with lip-sync mouth morph.
- Exit: session summary + KNOWN FACTS saved; Axerai AI + voice tokens logged per code.

2

Technology Stack (Complete)

LAYER	TECHNOLOGY	VERSION / NOTES
Language	JavaScript ES modules	.js / .jsx — no TypeScript
UI	React + Vite + Tailwind	19.x / 8.x / 4.x
3D / AR	Three.js + MindAR	0.184 / mind-ar image tracking
Character	FBX Mixamo rig + morph targets	D3 persona on AR anchor
AI	Google Gemini SDK	gemini-3.1-flash-lite primary
Voice TTS	Inworld API (doc)	Custom cloned persona; browser TTS fallback
STT	Web Speech API	en-IN — Roman Hinglish intent
Database	Supabase PostgreSQL	ledger_threads table
Hosting	Netlify	Static dist/ + Node 20 functions
Lint	ESLint 10	npm run lint

3 Repository Structure

```
axerai-love/
├─ index.html           SPA shell, mobile meta, viewport
├─ vite.config.js       React + Tailwind + three-mindar-shim
├─ tailwind.config.js
├─ netlify.toml         Build, Node 20, SPA redirect, functions/
├─ package.json
├─ scripts/verify-netlify-env.mjs  Pre-build secret leak guard
├─ public/
│  ├─ models/myra2.fbx   D3 character mesh (+ idle/talk FBX at runtime)
│  │  ├─ videos/target.mp4  Product card video on AR anchor
│  │  │  ├─ videos/intro.mp4  Optional intro media
│  │  │  └─ images/*-loading.png  Intro splash background
│  │  └─ targets.mind      MindAR image target (deploy asset)
│  └─ manifest.webmanifest
├─ netlify/functions/
│  ├─ gemini.mjs         Secure Gemini proxy
│  └─ elevenlabs-tts.mjs  TTS proxy (Inworld wiring in production doc)
├─ supabase/fresh_start.sql  Full schema + seed rows
└─ src/
   ├─ main.jsx           App vs AdminDashboard route split
   ├─ App.jsx            Main orchestrator (~3700 lines)
   ├─ AdminDashboard.jsx Password insights UI
   ├─ index.css          AR HUD, intro, live mic, admin styles
   ├─ apiProxy.js        /.netlify/functions/* fetch helpers
   ├─ supabaseClient.js  Anon Supabase client
   ├─ geminiModels.js    Model chains + flash/lite routing
   ├─ geminiUsage.js     Token normalization from API responses
   ├─ axeraiAssets.js    Weighted intro preload
   ├─ myraModel.js       D3 load, animations, mouth morph
   ├─ myraLipSync.js     Audio → mouth level 0-1
   ├─ myraPrompt.js      Persona system prompt + user prompts
   ├─ myraSummarize.js   Exit summary + KNOWN FACTS extraction
   ├─ myraLedger.js      Supabase memory, pairing, token logs
   ├─ myraVerify.js      Gemini 3-layer vision verify
   ├─ myraErrorFallback.js  Offline in-character error lines
   ├─ myraTargetVideo.js AR anchor product video plane
   ├─ myraStaticSession.jsx VR static scene (no camera)
   ├─ elevenLabsTts.js   TTS module (Inworld integration layer)
   ├─ mobileBrowser.js   iOS/Android detection
   ├─ mobileShell.js     Portrait lock, gesture block
   └─ three-mindar-shim.js  Three.js encoding shim
```

4 End-to-End User Flow

PHASE A — INTRO & PERMISSIONS

- User opens Netlify URL → `IntroShell` visible, `loadAxeraiExperienceAssets()` runs.
- Weighted preload: D3 FBX (5), target.mp4 (2), targets.mind (1), splash image (1). Min splash 1200ms.
- Progress bar reflects actual download — not fake timer.
- Camera grant on user gesture (iOS requirement). Back/environment camera preferred.
- Geolocation requested in background for regional slang in AI prompts.
- Intro handoff animation → `main-reveal` crossfade to AR viewport.

PHASE B — AR SESSION & LAYER A

- `MindARThree` starts — reuses intro camera stream when possible.
- User points at printed image target on packaging.
- **Layer A fires:** target found → anchor group live → product video on plane → D3 persona mounts.
- Persona may show after target video phase; verify runs in parallel (background).
- Torch toggle, camera flip available in HUD.

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User Flow (continued)

PHASE C — MEET CHARACTER & AUDIO GATE

- First TTS on iPhone requires explicit tap (“Meet character”) — unlocks Web Audio session.
- `needsAudioTap` gate; separate unlock audio element never overwrites TTS src.
- After unlock, subsequent lines auto-play.

PHASE D — LIVE CONVERSATION

- Default compose mode: **liveMic** — Web Speech API continuous recognition.
- Interim transcript in liquid capsule UI; “Tap to fix” before auto-send.
- Auto-send after `LIVE_MIC_SILENCE_MS = 2200` ms silence.
- While AI thinks or speaks: mic **fully released** (not muted); live mic panel hidden.
- Gemini reply → `prepareMyraSpeechText()` → Inworld TTS → lip sync + talk animation.
- `isTalking` true only on audio `onStart`, not when text arrives.
- Alternate modes: **keyboard** text + image upload; PTT fingerprint button in compose bar.

PHASE E — SESSION END

- User exits or `<SYSTEM_SLEEP>` token in Gemini reply triggers warm exit.
- `finishLedgerScan()` + `summarizeSessionDialogue()` via Gemini.
- KNOWN FACTS merged into locked memory block.
- Axerai AI tokens + voice tokens appended as JSON lines.
- Next scan injects memory — persona does not re-ask locked facts.

APP STATE MACHINE (SIMPLIFIED)

```
introVisible → asset preload → camera grant → intro exit
  → MindARSession → onTargetFound → target video → Layer A persona
  → runAnchorVerify (once, locked) → ledger pairing → welcome
  → liveMic|keyboard loop → exit → summarize → ledger save
Parallel: AR|VR toggle · verifyLocked · needsAudioTap · torch
```

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Intro Overlay & Asset Preload

File: `axeraiAssets.js` + `IntroShell` in `App.jsx`

ASSET	PATH	WEIGHT	PURPOSE
D3 FBX bundle	<code>preloadMyraModels()</code>	5	Character + idle + talk clips
Target video	<code>/videos/target.mp4</code>	2	Product card on AR anchor
MindAR target	<code>/targets.mind</code>	1	Image tracking definition
Splash art	<code>/images/*-loading.png</code>	1	Branded loading background

- `MIN_SPLASH_MS = 1200` — minimum branded splash even if assets cache-fast.
- Camera permission UI: tap ring, retry card (`.intro-access-*` CSS).
- iOS strategy: `getUserMedia` on first pointerdown before audio/geo prompts.
- Failed preload surfaces error state; user can retry grant.
- `logAxeraiBuildConfig()` logs env fingerprint on App mount (dev debug).

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AR / MindAR Configuration

SETTING	VALUE / BEHAVIOR
Target file	<code>/targets.mind</code>
Library	<code>MindARThree</code> from mind-ar
UI chrome	uiLoading/uiScanning/uiError: 'no' — custom HUD only
Camera	Reuse intro stream via startMindARWithPreviewStream()
Anchor	Image target 0 → anchorGroup → video plane + D3 wrapper
Lighting	Ambient 1.4 + directional 2.5 + fill 1.2
Character scale	0.008 · position tuned to product anchor
Scan guide	HUD: “Center the product / brand card”
Teardown	softTeardownMindAR() keeps camera alive between sessions
VR mode	myraStaticSession.jsx — static art, no camera, same chat stack

TARGET VIDEO PHASE

File: `myraTargetVideo.js`

- MP4 on anchor plane before or during persona reveal.
- 16s watchdog + Safari inline playback hacks.
- `TARGET_VIDEO_SAFETY_MS` force-advances stuck Safari playback.
- `targetVideoDone` gates welcome message scheduling.

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Three-Layer Verification

Product logic (patent): Layer A = fast visibility engine. Persona appears when product/target visible. Layers B + C run in background; return true/false without blocking Stage 1.

A · FAST

Product Visibility Engine
Trigger: MindAR target found / product in frame.
Action: Instant 3D persona on anchor (Stage 1).
Implementation: MindAR onTargetFound — not a Gemini call.

B · BG

Code / Brand Verification
Trigger: After Layer A — background frame capture.
Action: Verify brand text / product code on packaging (e.g. AXE-327T5).
Output: true = authentic · false = wrong/missing/unreadable.

C · BG

Liveness / Anti-Spoof
Trigger: Parallel with B.
Action: Detect real product vs photo-on-screen, monitor replay, print spoof.
Output: true = live product · false = spoof.

IMPLEMENTATION MAPPING (GEMINI VISION — MYRAVERIFY.JS)

GEMINI LAYER	MAPS TO	CHECKS
LAYER1 brand text	Product Layer B	Full brand name visible — no partial guess
LAYER2 liveness	Product Layer C	Real card in hand vs screen/photo spoof
LAYER3 frame	Quality gate	Card centered, readable, not random background

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Verification Implementation (continued)

SINGLE GEMINI CALL — STRUCTURED OUTPUT

```
LAYER1: PASS | FAIL
LAYER2: PASS | FAIL
LAYER3: PASS | FAIL
RESULT: REAL | PHOTO_SPOOF | NO_BRAND | BAD_FRAME | UNCERTAIN

REAL = all PASS → code assigned (e.g. AXE-327T5)
PHOTO_SPOOF = LAYER2 FAIL (priority over others)
NO_BRAND = LAYER1 FAIL
BAD_FRAME = LAYER1 PASS + LAYER3 FAIL
```

PIPELINE

- Trigger: mindarReady && showMindAR && !verifyLocked && !isVerified
- Canvas snap from MindAR video → compressImageForVerify (max 768px JPEG 0.82)
- Gemini via proxy · temperature 0 · VERIFY_MODEL_CHAIN · 1 retry
- `verifyLockedRef` — one attempt per AR session (pass or fail)
- Verify fail: HUD note + in-character error via myraErrorFallback (not saved to ledger)
- Persona visibility: continues after target video even if verify fails; verify gates ledger pairing

FAIL REASONS → UX

RESULT	SITUATION KEY	USER EXPERIENCE
PHOTO_SPOOF	SCAN_PHOTO_SPOOF	Persona line: not a real product
NO_BRAND	SCAN_CARD_NOT_FOUND	Brand/code not readable
BAD_FRAME	SCAN_BAD_FRAME	Re-center product in frame
UNCERTAIN	SCAN_GLITCH	Try again messaging
Pair full	SCAN_PAIR_FULL	3rd device blocked — 2 roles filled

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D3 Character — Model & Animation

ASSET	PATH	PURPOSE
Character mesh	/models/myra2.fbx	Skinned D3 persona
Idle	idle.fbx	Standing loop when silent
Talk clips	talking-4/5/6.fbx	Random while speaking
Placement	pos (0,-0.35,-0.15) scale 0.008	Tuned to packaging anchor

- Three.js AnimationMixer on skinned mesh.
- Idle ↔ Talk crossfade 0.35s — no hard stop (prevents T-pose flash).
- Random talk clip; avoids same clip twice consecutively.
- preloadMyraModels() + SkeletonUtils.clone() per session.
- tickMyraMixer(anchorGroup, delta) every animation frame.

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Lip Sync Pipeline

STEP 1 — AUDIO ANALYSIS (MYRALIPSYNC.JS)

- TTS <audio> → Web Audio AnalyserNode → mouth level 0 (closed) to 1 (open).
- Mid-frequency speech band energy mapped to mouth openness.
- iPhone: skip Web Audio graph → procedural sine mouth (protects audio session).
- Smoothed per frame; eases closed on speech end.

STEP 2 — MESH APPLICATION (MYRAMODEL.JS)

- **Priority A:** morph target `mouth open` (also mouthopen, jaw, lip keys).
- **Priority B:** head bone micro-tilt if no morph targets.
- Smoothing factor 0.32 per frame on morph influence.
- connectTtsAudio() / disconnectTtsAudio() on TTS lifecycle.

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AI Brain — Google Gemini

MODEL CHAINS (GEMINIMODELS.JS)

USE	PRIMARY	FALLBACK
Chat	gemini-3.1-flash-lite	gemini-flash-lite-latest
Verify	same chain	same
Summary	same chain	same

FLASH VS LITE TIER ROUTING

- Flash tier (temp 0.95): first 6 user turns, messages ≥ 150 words, welcome/boot/resume.
- Lite tier (temp 0.85): default mid-session chat.
- resolveMyraChatModels() returns models + tier + reason string for logging.

PERSONA PROMPT (MYRAPROMPT.JS)

- Large MYRA_SYSTEM_PROMPT — Axerai Core Directive + character bible.
- Output: casual Hinglish Roman script; emotion-first; no bot/API jargon.
- Scenarios: retail browse, keepsake gift, sender/receiver gift flows.
- buildMyraUserPrompt({ type: welcome|resume|reply|boot }) per turn.
- KNOWN FACTS (LOCKED) block injected from ledger — no re-ask name/gift-for/occasion.
- Inworld path: sparse audio tags [laughs softly] stripped from ledger text.

CALL TYPES LOGGED

CALLTYPE	WHEN
verify	3-layer vision frame check
welcome	First greeting after scan
chat	User message reply
summary	Session exit summarization

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Voice — Inworld API

MVP documentation voice: Inworld TTS with custom cloned brand persona. Integration module in codebase: `elevenLabsTts.js` + `elevenlabs-tts.mjs` proxy. Browser speechSynthesis is fallback.

CONFIGURATION

ENV (LOCAL)	PURPOSE
VITE_INWORLD_ENABLED=true	Enable cloud voice
VITE_INWORLD_VOICE_ID	Cloned persona voice ID

ENV (NETLIFY SERVER)	PURPOSE
INWORLD_API_KEY	Server-side only — never in browser bundle
INWORLD_VOICE_ID	Default voice for proxy

TTS FLOW

- 1. Gemini text → prepareMyraSpeechText() — strip markdown/emojis.
- 2. PROD: POST /.netlify/functions/inworld-tts (proxy) → MP3 blob.
- 3. Hidden <audio> playback → connectTtsAudio() → lip sync.
- 4. Response header X-Axera-Characters → voice token log.
- 5. Failure → pickMyraVoice() browser speechSynthesis (Swara/hi/en-IN priority).
- 6. MYRA_TTS_SAFETY_MS = 35000 — recovery if TTS hangs.

IPHONE / SAFARI CRITICAL RULES

- Separate unlock audio element — never overwrite active TTS src.
- First line deferred until Meet-character tap (user gesture).
- Silent oscillator keep-alive on iOS only.
- Mic fully released during TTS — prevents echo + Android STT conflict.
- primeSafariSpeechSynthesis() + unlockMobileSpeechAudio() on first touch.

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Speech-to-Text & Input Modes

SETTING	VALUE
API	Browser Web Speech API
Language	en-IN (Roman Hinglish intent — avoid hi-IN Devanagari)
Auto-send silence	LIVE_MIC_SILENCE_MS = 2200
Voice energy threshold	LIVE_MIC_VOICE_ENERGY = 20
Vision upload	GEMINI_VISION_ENABLED = false (UI exists, disabled)

TAP TO FIX

- Transcript tappable → edit mode with textarea + Send / Cancel.
- Auto-send timer paused while editing; STT onresult ignored.
- Cancel restores silence timer or restarts mic.

COMPOSE MODES (APP.JSX)

MODE	UI	BEHAVIOR
liveMic (default)	Liquid capsule + voice visualizer	Continuous STT → auto-send
keyboard	Text field + optional image	Manual send; PTT fingerprint in bar
HUD toggle	Keyboard / mic icons	Switch modes mid-session

ANDROID MIC RULE

Do not hold getUserMedia for level meter while STT active — steals mic from recognition. Use procedural level animation instead.

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Supabase Ledger & Memory

Table: `ledger_threads` — max 2 rows per product code (sender + receiver). Schema: `supabase/fresh_start.sql`

COLUMN	TYPE	PURPOSE
verification_code	text	Product code e.g. AXE-327T5
device_id	text	Stable browser ID (localStorage + 5yr cookie)
role	text	sender receiver
scan_count	int	Completed scans on this thread
conversation	text	Full chat with session markers
session_summaries	text	Exit summaries + FACTS per scan
axerai_ai_usage	text	JSON lines — Gemini tokens
axerai_voice_usage	text	JSON lines — Inworld characters

DEVICE PAIRING RULES

- First device on code → sender thread. Second distinct device → receiver.
- Third device when both roles filled → SCAN_PAIR_FULL (blocked).
- `resolveSessionAccess()` determines role before welcome.
- Welcome modes: SENDER_FIRST, SENDER_RETURN, RECEIVER_FIRST, RECEIVER_RETURN.

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Session Summaries & KNOWN FACTS

File: `myraSummarize.js`

- Exit: `summarizeSessionDialogue()` via Gemini — FACTS / STORY / OPEN HOOK / BRAND PRAISE format.
- `extractFactsFromSummary()` → `mergeKnownFacts()` → `formatKnownFactsBlock()`.
- `buildGeminiMemoryText()` injects locked facts + past story into next session prompt.
- Offline `myraErrorFallback` lines NEVER saved to ledger.
- Session markers: --- session N start/end --- in conversation column.

USAGE JSON LINE EXAMPLES

```
{ "at": "ISO8601", "call": "chat", "model": "gemini-3.1-flash-lite", "prompt": 4200, "output": 180, "total": 4380 }
{ "at": "ISO8601", "call": "tts", "model": "inworld-tts", "voice": "persona-id", "chars": 287 }
```

15 Admin Dashboard

File: `AdminDashboard.jsx` · **Route:** `/VITE_DASHBOARD_PATH` (default `axerai-insights-7k2m`)

- Routed in main.jsx by pathname or hash — skips mobile shell init.
- Password gate: VITE_DASHBOARD_PASSWORD → sessionStorage auth.
- Per-code filter (default seed: AXE-327T5 in doc / pilot codes in DB).
- Panels: scan analytics, conversation bubble parse, duration formatting.
- buildGeminiUsageAnalytics() — totals by call type, per-entry table.
- buildElevenLabsUsageAnalytics() — voice character totals (Inworld in doc).
- Brand product praise parsed from session summaries.
- Refresh + logout controls.

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Environment Variables

LOCAL DEVELOPMENT (.ENV)

```
VITE_GEMINI_API_KEY=      # Direct Gemini in dev
VITE_INWORLD_ENABLED=true
VITE_INWORLD_VOICE_ID=
VITE_SUPABASE_URL=
VITE_SUPABASE_ANON_KEY=
VITE_DASHBOARD_PATH=axerai-insights-7k2m
VITE_DASHBOARD_PASSWORD=
```

NETLIFY PRODUCTION (SERVER — NOT IN BROWSER BUNDLE)

```
GEMINI_API_KEY=          # Required — build fails without
INWORLD_API_KEY=         # Inworld TTS server key
INWORLD_VOICE_ID=
NODE_VERSION=20
VITE_SUPABASE_URL=       # OK in client
VITE_SUPABASE_ANON_KEY=  # OK in client
VITE_DASHBOARD_PATH=
VITE_DASHBOARD_PASSWORD=
```

Build guard: verify-netlify-env.mjs blocks VITE_GEMINI_API_KEY and VITE_INWORLD_API_KEY on Netlify — secrets must use server keys only.

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Netlify Deployment & API Endpoints

ITEM	VALUE
Build cmd	node scripts/verify-netlify-env.mjs && npm run build
Publish	dist/
Node	20
SPA	/* → /index.html status 200

ENDPOINT	METHOD	BODY / RESPONSE
/netlify/functions/gemini	GET	Health / key fingerprint
/netlify/functions/gemini	POST	userPrompt, systemInstruction, models[], imagePart?, generationConfig
/netlify/functions/inworld-tts	POST	text, voiceId? → MP3 + X-Axera-Characters header

Local dev: USE_API_PROXY false when not PROD — direct SDK with VITE_GEMINI_API_KEY.

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UI / CSS Components (index.css)

CLASS PREFIX	PURPOSE
.intro-* / .intro-access-*	Intro shell, camera permission tap ring
.myra-live-mic-*	Live mic panel, capsule, voice energy (~230 lines)
.myra-live-mic-edit	Tap-to-fix edit UI
.hud-*	AR overlay: scan guide, verify note, side dock, AR/VR toggle
.axerai-audio-tap	Meet character iOS unlock button
.compose-* / .myra-ptt-*	Keyboard compose + fingerprint PTT
.admin-dash_*	Dashboard layout and analytics tables
.main-reveal	Intro → AR crossfade animation

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Error Handling Catalog

File: `myraErrorFallback.js`

PHASE	SITUATIONS
SCAN	SCAN_CARD_NOT_FOUND, SCAN_PHOTO_SPOOF, SCAN_BAD_FRAME, SCAN_GLITCH, SCAN_PAIR_FULL, SCAN_MAGIC_ASLEEP
WELCOME	WELCOME_MAGIC_OFF, WELCOME_QUOTA, WELCOME_CONNECTION_WEAK, WELCOME_BUSY, WELCOME_GLITCH
CHAT	CHAT_QUOTA, CHAT_CONNECTION_WEAK, CHAT_BUSY, CHAT_GLITCH
INPUT	NO_SPEECH, MIC_BLOCKED, PHOTO_FAIL

- `pickMyraErrorLine(situation)` — random in-character Hinglish; no repeat same session.
- `classifyGeminiError()` maps API errors to situations.
- Silent errors (no TTS): `NO_SPEECH`, `CHAT_GLITCH`, `CHAT_QUOTA`.
- `MYRA_ERROR_REPEAT_RISK`: high/medium/low/once per situation.

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Key Constants & State Machine

CONSTANT	VALUE	MEANING
MINDAR_TARGET	/targets.mind	AR target file
SPEECH_RECO_LANG	en-IN	STT language
LIVE_MIC_SILENCE_MS	2200	Auto-send delay
LIVE_MIC_VOICE_ENERGY	20	Mic level threshold
MYRA_TTS_SAFETY_MS	35000	Max TTS wait before recovery
GEMINI_VISION_ENABLED	false	User photo to Gemini in chat off
GEMINI_RETRIES_PER_MODEL	2	API retry count
MYRA_FLASH_USER_TURNS	6	Flash tier turn count
MIN_SPLASH_MS	1200	Intro minimum display

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Security, VR Mode, Mobile Shell

SECURITY MODEL

- 3-layer verify + one cycle per session.
- API keys server-side in production.
- Supabase RLS: anon insert/select/update on ledger_threads (pilot — tighten at scale).
- Dashboard: secret path + password, sessionStorage only.
- Device ID = continuity, not authentication.
- Per-code usage caps (pilot policy) — soft exit ~200–300 voice tokens remain.
- Patent-pending multi-layer verification architecture.

VR MODE & MOBILE SHELL

- experienceViewMode: ar | vr — toggle in HUD.
- VR: MyraStaticSession — static brand art background, same chat/voice stack.
- mobileShell.js: portrait orientation lock, block pinch/double-tap zoom, select prevention.
- mobileBrowser.js: iOS detection drives lip sync + audio unlock paths.

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Per-Code Budget, Build & Not in MVP

PILOT BUDGET PACKAGES (EXAMPLE AXE-327T5)

PACKAGE	TIME	AI TOKENS	VOICE TOKENS
Light	~12 min	~59,000	~4,000
Standard	~30 min	~1.5L	~10,000
Deep	~60 min	~2.9L	~20,000

Policy: when ~200–300 voice tokens remain → persona warm exit line → session ends for that code. (Pricing model — enforcement configurable per pilot.)

BUILD & RUN

```
npm install
npm run dev      # localhost Vite
npm run build    # production dist/
npm run preview
npm run lint

Fresh DB: Supabase SQL Editor → run supabase/fresh_start.sql only
```

NOT IN MVP YET

- Self-serve brand dashboard (upload persona, register products).
- Inworld STS full speech-to-speech pipeline.
- Google Cloud STT (browser only today).
- Own GPU inference server (future scale path documented in pricing docs).
- Multi-tenant brand admin.
- Native app store application.

One-line definition: Axerai MVP = scan packaging → Layer A instant persona → background B/C verify → Gemini conversation with ledger memory → Inworld voice with lip-sync → Axerai AI + voice tokens per code `AXE-327T5` .

End of Full Technical Reference. Companion overview: AXERAI-MVP-Document.pdf. No application code modified by this document.